

Enterprise Network Design and Access Control List (ACL) Implementation

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1. Introduction

This project involved designing and implementing a small enterprise network in Cisco Packet Tracer. The network consists of three departments (Sales, HR and IT), each connected through a dedicated switch to a Cisco router. Static IP addressing was configured for all hosts and inter-network communication was enabled. An Extended Access Control List (ACL) was then implemented to prevent the HR department from accessing the IT subnet while allowing legitimate communication from the Sales department.

2. Network Topology

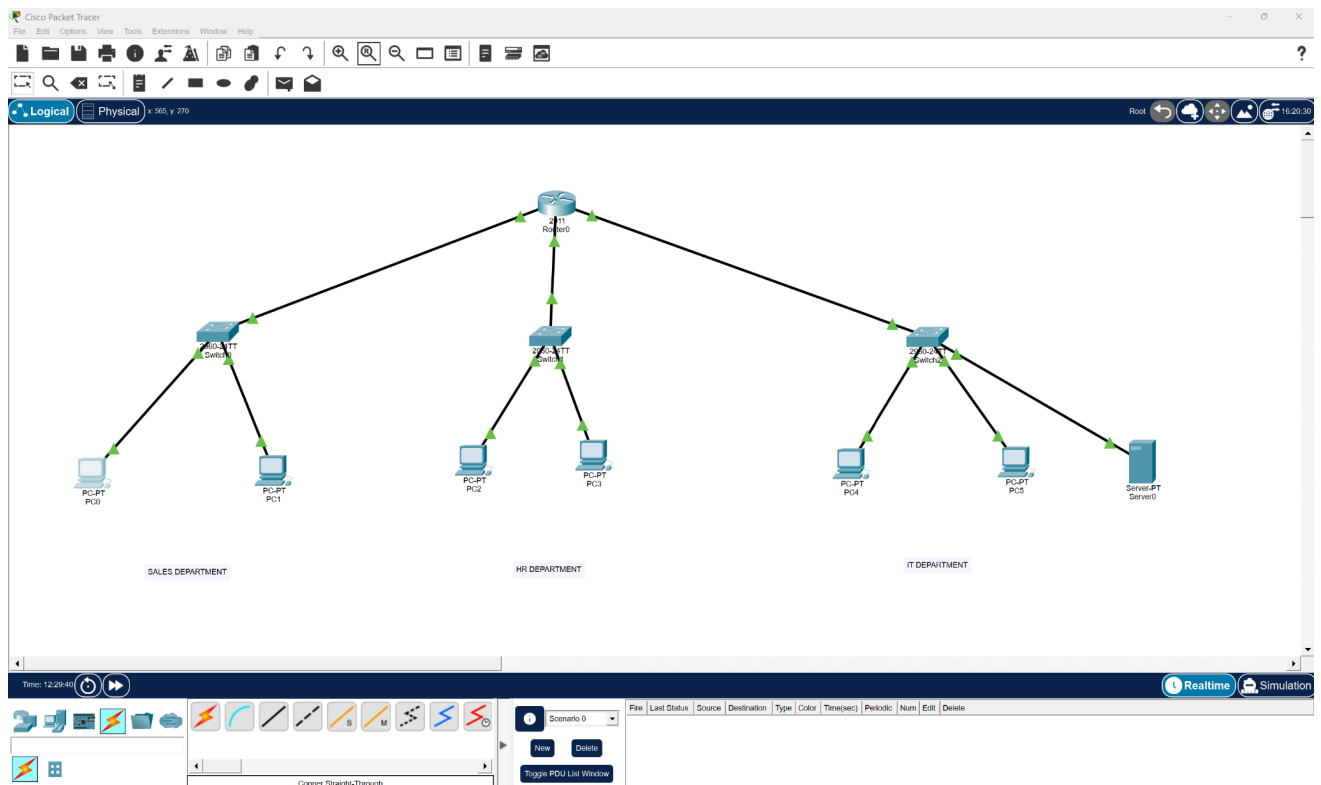


Figure 1: Completed Enterprise Network Topology

3. IP Address Configuration

The following screenshots show the static IP configuration for the devices used in each department.

The screenshot shows a configuration window for PC0 with tabs for Physical, Config, Desktop, Programming, and Attributes. The Desktop tab is active, displaying the IP Configuration section for the FastEthernet0 interface. The configuration is set to Static, with the following values:

Field	Value
Interface	FastEthernet0
IP Configuration	
☐ DHCP	☒ Static
IPv4 Address	192.168.20.10
Subnet Mask	255.255.255.0
Default Gateway	192.168.20.1
DNS Server	0.0.0.0
IPv6 Configuration	
☐ Automatic	☒ Static
IPv6 Address	
Link Local Address	FE80::201:C7FF:FEA9:D777
Default Gateway	
DNS Server	
802.1X	
☐ Use 802.1X Security	
Authentication	MD5
Username	
Password	

At the bottom left, there is a checkbox labeled "Top".

Figure 2: Sales PC0 IP Configuration

PC2

Physical Config **Desktop** Programming Attributes

IP Configuration [X]

Interface: FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address: 192.168.30.10

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.30.1

DNS Server: 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address: /

Link Local Address: FE80::250:FFF:FE5D:7C41

Default Gateway:

DNS Server:

802.1X

☐ Use 802.1X Security

Authentication: MD5

Username:

Password:

☐ Top

Figure 3: HR PC2 IP Configuration

PC4

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.10.2

Subnet Mask 255.255.255.0

Default Gateway 192.168.10.1

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::290:21FF:FE99:6736

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

☐ Top

Figure 4: IT PC4 IP Configuration

The screenshot shows a configuration window for 'Server0' with tabs for Physical, Config, Services, Desktop, Programming, and Attributes. The 'Config' tab is active, and a sub-tab 'IP Configuration' is selected. The window contains three main configuration sections:

- IP Configuration:** Features radio buttons for 'DHCP' and 'Static' (selected). Below are input fields for IPv4 Address (192.168.10.10), Subnet Mask (255.255.255.0), Default Gateway (192.168.10.1), and DNS Server (0.0.0.0).
- IPv6 Configuration:** Features radio buttons for 'Automatic' and 'Static' (selected). Below are input fields for IPv6 Address (empty), Link Local Address (FE80::206:2AFF:FE03:92BC), Default Gateway (empty), and DNS Server (empty).
- 802.1X:** Includes a checkbox for 'Use 802.1X Security' (unchecked). Below are a dropdown for 'Authentication' (set to MD5), and input fields for 'Username' and 'Password'.

A 'Top' button is located at the bottom left of the window.

Figure 5: Server IP Configuration

4. Connectivity Verification

Connectivity tests were performed using the ping command. The Sales department successfully communicated with the IT network, confirming proper routing. After implementing the ACL, the HR department was unable to communicate with the IT subnet, confirming that the security policy was successfully enforced.

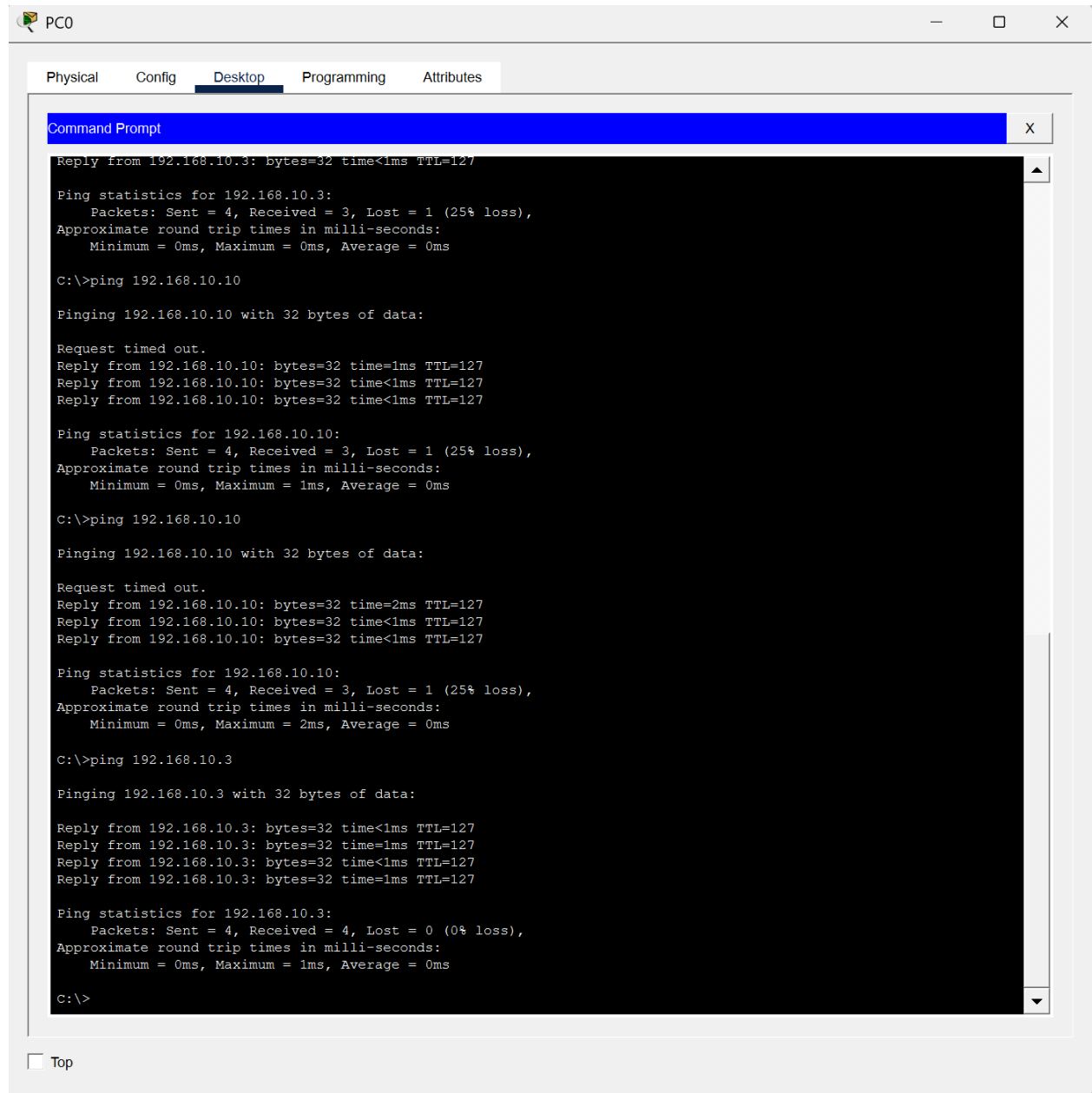


Figure 6: Successful Ping from Sales PC to IT Network

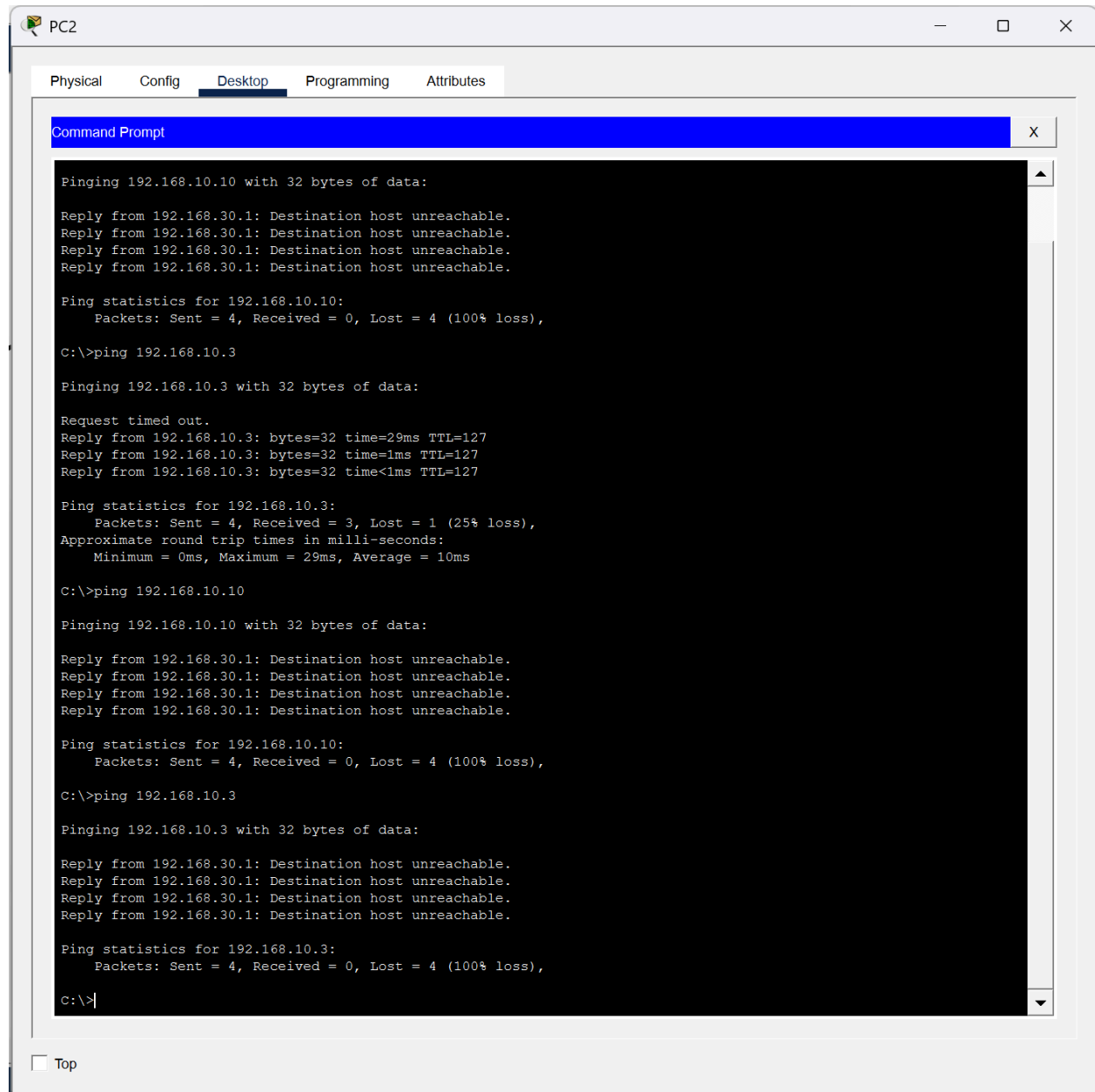


Figure 7: Failed Ping from HR PC to IT Network

5. Router Configuration and ACL

The router interfaces were configured with gateway addresses for each subnet. An Extended Access Control List (ACL 100) was configured and applied inbound on the HR interface to block all traffic originating from the HR subnet (192.168.30.0/24) to the IT subnet (192.168.10.0/24), while permitting all other traffic.

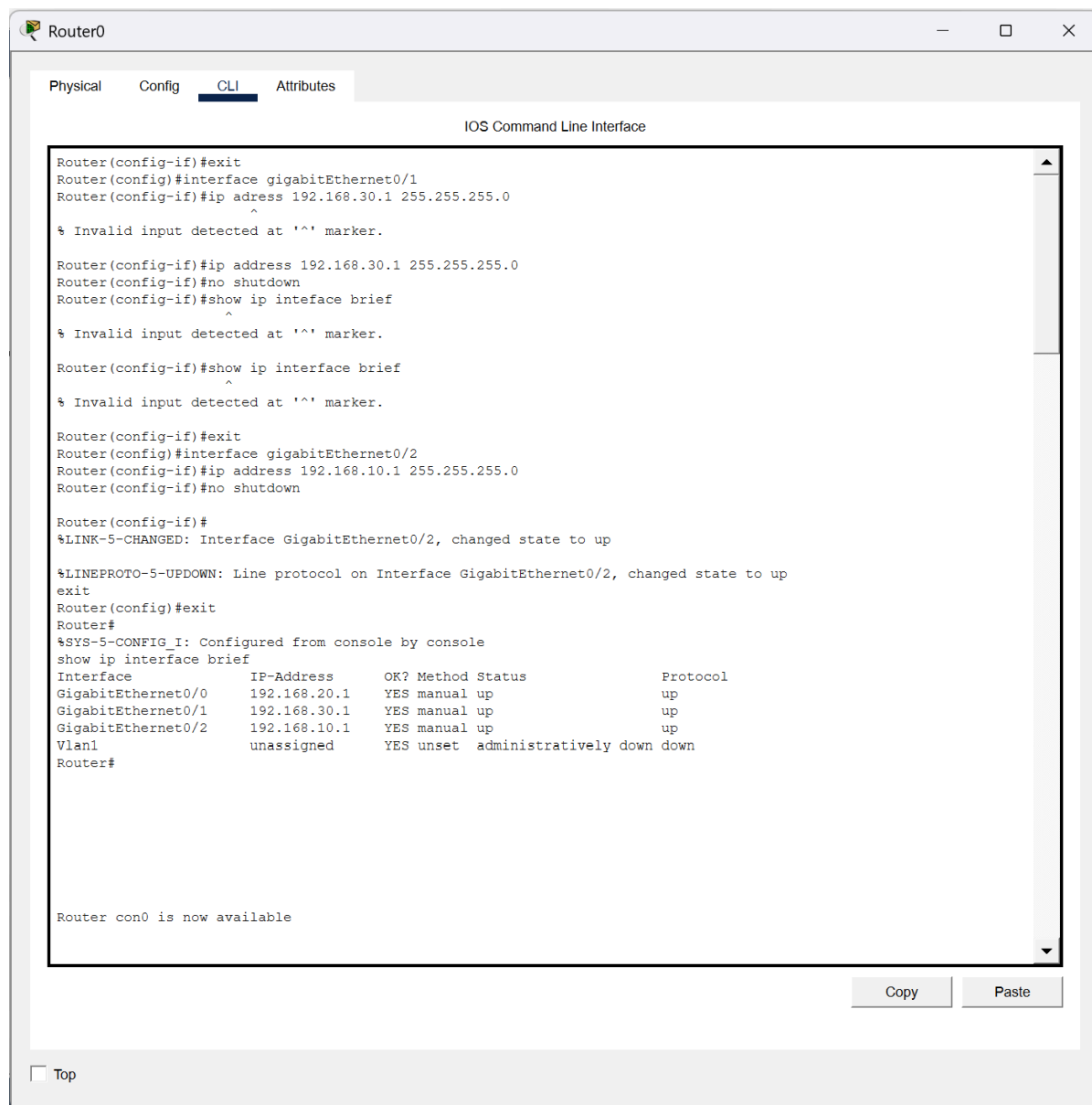


Figure 8: Router Interface Configuration

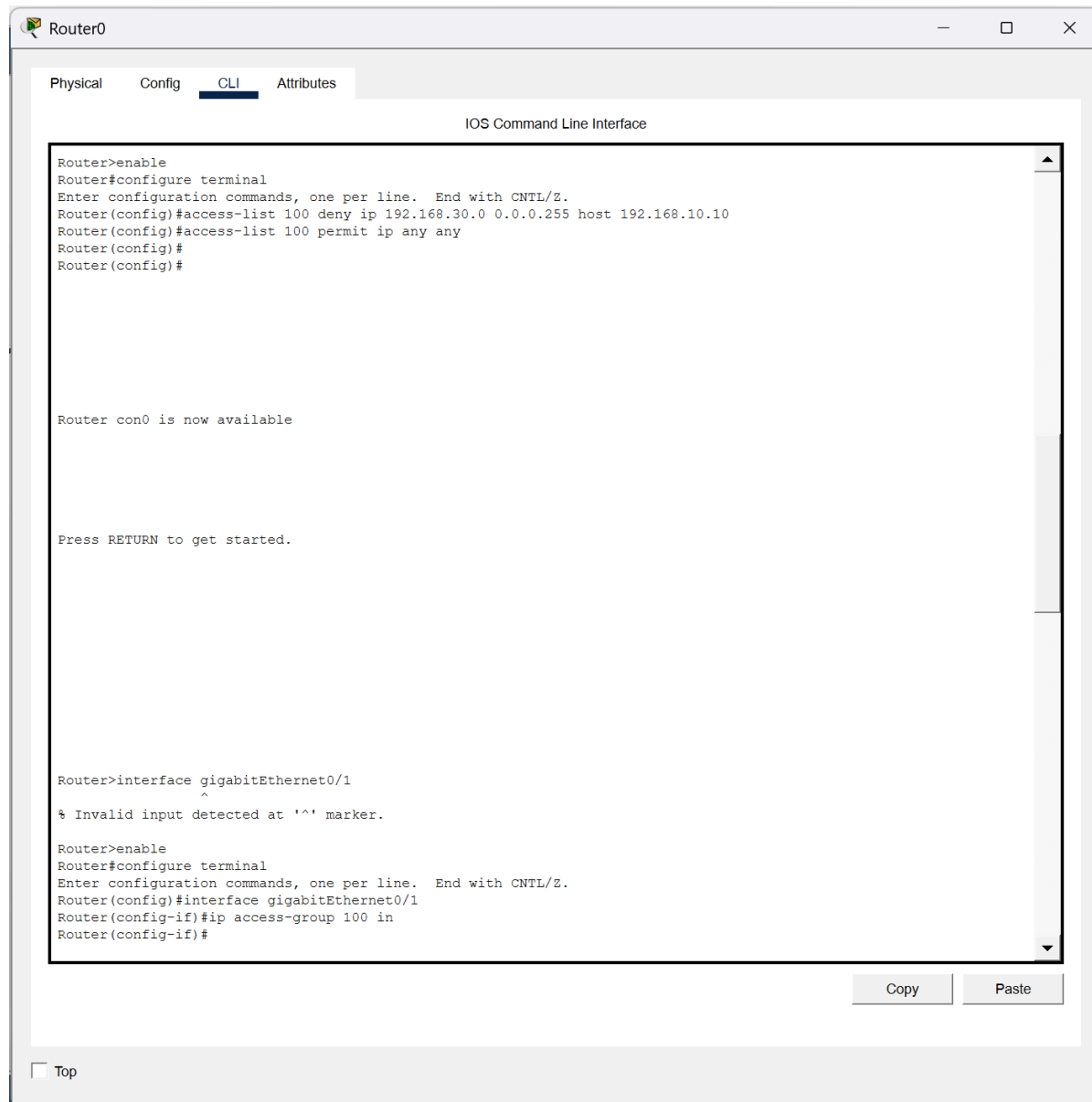


Figure 9: ACL Applied to HR Interface

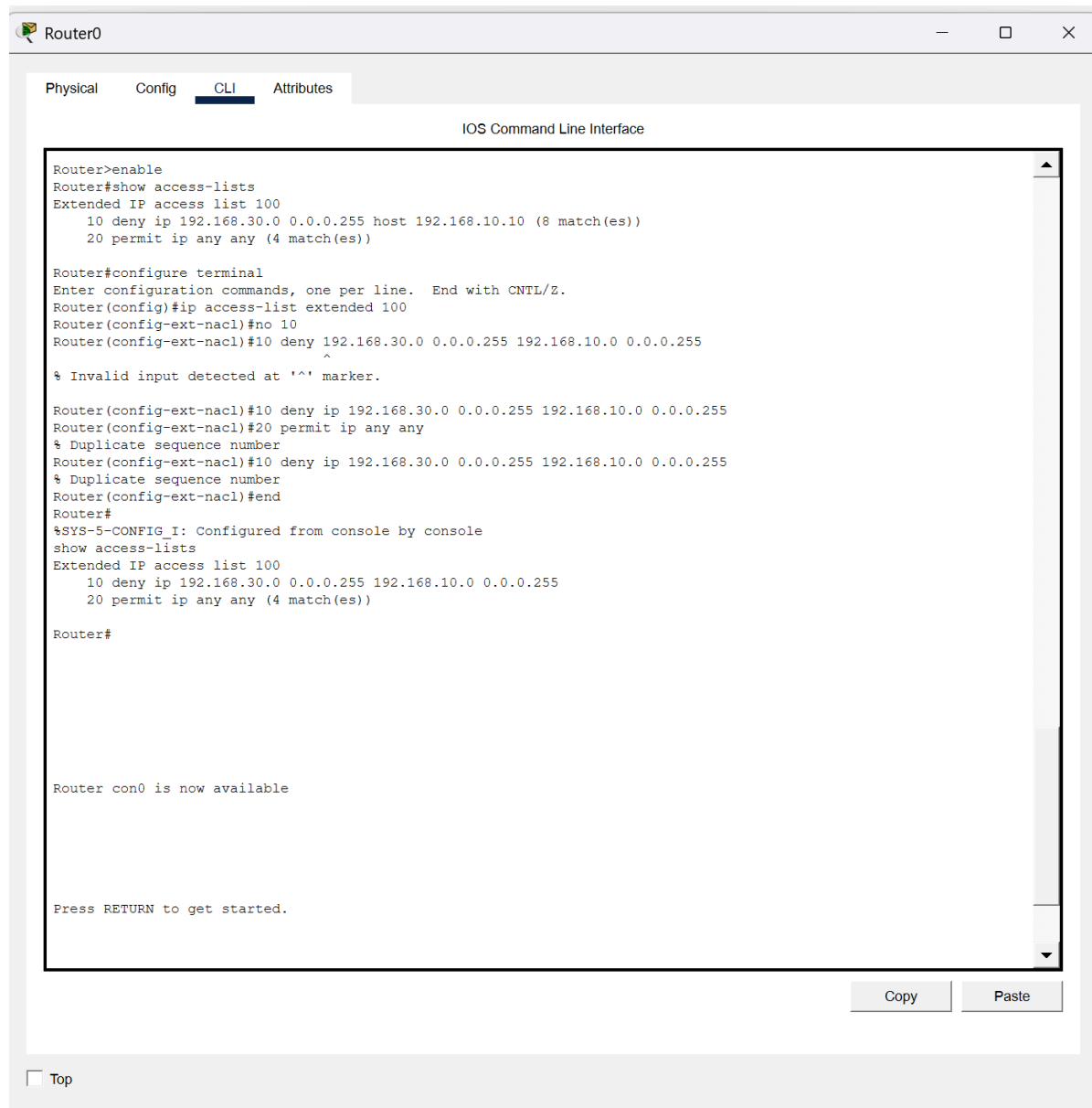


Figure 10: Final ACL Verification

6. Results

Test	Expected Result	Outcome
Sales PC → IT PC	Successful communication	Passed
Sales PC → Server	Successful communication	Passed
HR PC → IT PC	Communication blocked	Passed
HR PC → Server	Communication blocked	Passed

7. Conclusion

The objectives of the assignment were successfully achieved. The enterprise network was designed and configured using Cisco Packet Tracer, static IP addressing was implemented, routing between the three departments was verified, and an Extended ACL was successfully deployed to restrict HR users from accessing the IT subnet while maintaining authorised communication for other departments. The screenshots included in this report provide evidence that all required tasks were completed successfully.